REVIEW

CURRENT OPINION

# Intraoperative mechanical ventilation during thoracic surgery: contemporary controversies and evolving strategies

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**Purpose of review**
To investigate and synthesize recent evidence on intraoperative mechanical ventilation strategies during thoracic surgery, with a focus on the controversies of fixed versus individualized positive end-expiratory pressure (PEEP), high versus low PEEP, mechanical and chemical power, driving pressure, and the integration of therapy bundles.

**Recent findings**
Recent randomized controlled trials and meta-analyses have claimed that individualized PEEP titration reduces postoperative pulmonary complications (PPCs) compared with fixed PEEP. High PEEP does not universally confer benefit and may increase hemodynamic risk, while low PEEP risks atelectasis. Driving pressure has emerged as a potential modifiable risk factor for PPCs, with individualized strategies to minimize driving pressure showing improved outcomes. Mechanical power, is increasingly recognized as a predictor of PPCs, while chemical power is emerging as a new concept. Therapy bundles, including perioperative open-lung approaches and individualized postoperative care, are promising but require further validation.

**Summary**
The current literature shows a trend toward individualized perioperative ventilation strategies in thoracic surgery, particularly individualized PEEP titration. Therapy bundles and individualized postoperative care, including high-flow nasal oxygen, are emerging as adjuncts to reduce PPCs. Large-scale, definitive trials are needed to determine the clinical value of these strategies and clarify the role of chemical power and therapy bundles.

**Keywords**
mechanical power, one-lung ventilation, postoperative pulmonary complications, thoracic anesthesia

## INTRODUCTION

Intraoperative mechanical ventilation during thoracic surgery is a field marked by rapid evolution and ongoing debate. The unique physiological challenges of thoracic procedures – particularly the frequent use of one-lung ventilation (OLV) – dictate careful titration of ventilatory parameters to minimize the risk of postoperative pulmonary complications (PPCs), which remain a leading cause of morbidity and mortality in this patient population [1].

The use of tidal volumes of approximately 5 ml/kg of predicted body weight is widely recognized as a cornerstone of lung-protective OLV. Although not entirely undisputed [2], this approach is supported by both clinical guidelines and robust meta-analyses [3▪,4]. In contrast, other aspects of OLV have seen a shift from standardized protocols toward more individualized strategies.

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## KEY POINTS

- High positive end-expiratory pressure (PEEP) does not universally improve outcomes and may increase hemodynamic risk; low PEEP risks atelectasis and hypoxemia.
- Driving pressure is a central modifiable risk factor for PPCs; strategies to minimize driving pressure improve intraoperative oxygenation and may reduce complications.
- Mechanical power, particularly its driving pressure component, is associated with postoperative respiratory failure.
- Therapy bundles and individualized postoperative care, including high-flow nasal oxygen, are promising but require further research for routine adoption.

Particular attention has been given to optimizing positive end-expiratory pressure (PEEP), understanding the implications of mechanical power and energy delivery to the lungs, as well as perioperative therapy bundles. These bundles include protective OLV strategies, combined with early mobilization and physiotherapy, incentive spirometry, and the application of postoperative noninvasive ventilation or high-flow oxygen therapy (HFNO) [5]. Despite the strong physiological rationale and technological advancements supporting these personalized approaches, recent high-quality evidence – including large randomized controlled trials (RCTs) and meta-analyses – has called into question the universal benefit of such individualized interventions [6▪,7]. These findings suggest that while tailoring ventilatory settings may benefit selected patient populations, a broad application of these strategies does not consistently yield improved outcomes across all clinical scenarios [6▪–8].

This review synthesizes the most recent evidence and addresses the key controversies in intraoperative mechanical ventilation during thoracic surgery. Special attention is given to the debates on individualized versus fixed PEEP, the clinical significance of mechanical and chemical power and cumulative mechanical energy, and the integration of postoperative care strategies such as HFNO. The aim is to provide anesthesiologists and perioperative teams with a state-of-the-art perspective regarding optimization of intraoperative ventilation to minimize harm and improve recovery in thoracic surgical patients. Key aspects of protective OLV are summarized in Fig. 1.

# MAIN TEXT

## Positive end-expiratory pressure

The application of PEEP during mechanical ventilation is intended to stabilize lung units, improve oxygenation, and reduce the risk of atelectrauma by maintaining end-expiratory lung volume [9]. In thoracic surgery, especially during OLV, the risk of alveolar derecruitment and subsequent PPCs is increased because of the nonphysiological ventilation of a single lung and the frequent use of high inspired oxygen fractions. The concept of individualized PEEP – PEEP titrated to optimize respiratory mechanics, ventilation and perfusion distribution, or gas exchange – arose from the recognition that fixed PEEP levels may be either insufficient or too excessive, depending on patient-specific lung and chest wall characteristics, surgical positioning, and intraoperative events. Meta-analyses and RCTs have demonstrated that individualized PEEP, particularly when titrated to maximal dynamic compliance or minimal driving pressure, can improve intraoperative oxygenation, reduce driving pressure, and enhance respiratory mechanics during OLV [7,10▪]. A recent 2025 meta-analysis of 10 RCTs with a total of more than 3400 patients found that individualized PEEP titration by lung compliance was associated with a significant reduction in the composite risk of PPCs compared with fixed PEEP [risk ratio: 0.55, 95% confidence interval (CI): 0.38–0.78] [7]. Individualized PEEP also reduced the risk of specific complications such as pneumonia and atelectasis, and improved oxygenation indices and dynamic compliance without increasing hemodynamic instability. The physiological rationale is compelling; by tailoring PEEP to the individual's lung mechanics, clinicians may avoid both derecruitment and overdistension [11].

Despite these promising physiologic and surrogate outcome improvements, recent high-quality evidence has cast doubt on the universal clinical benefit of individualized PEEP in reducing PPCs across broader patient and surgical populations. The PRIME-AIR study, a large multicenter RCT, evaluated a comprehensive perioperative lung expansion bundle – including repeated individualized PEEP titration to maximize respiratory system compliance – against usual care in patients undergoing major open abdominal surgery [8]. Despite high adherence to the intervention and improved intraoperative respiratory mechanics, the study found no significant reduction in the severity or incidence of PPCs compared with usual care (mean difference in PPCs severity grade 0.07, 95% CI: –0.03 to 0.18; $P = 0.19$). Notably, the intervention group received higher mean PEEP (7.5 $cmH_2O$) than the control

 

**Tidal Volume**

- ✓ Set tidal volume to 4–5 mL/kg predicted body weight and up to 6 mL/kg if Driving Pressure is low

**PEEP (Positive End-Expiratory Pressure)**

- ✓ Apply PEEP of 5–10 $cmH_2O$, individualized to patient's needs including oxygenation and hemodynamic status
- ✓ Consider alveolar recruitment maneuvers in case of hypoxemia: Perform alveolar recruitment with the ventilator; avoid manual recruitment

**Driving Pressure** (ΔP = plateau pressure – PEEP)

- ✓ Monitor and minimize driving pressure , ideally <12–15 $cmH_2O$
- ✓ Adjust ventilator settings to keep ΔP as low as possible

**$F_IO_2$ (Fraction of Inspired Oxygen)**

- ✓ Use the minimum $F_IO_2$ necessary to maintain $SpO_2$ >90%
- ✓ Titrate down from 1.0 as soon as feasible to avoid oxygen toxicity

**Permissive Hypercapnia**

- ✓ Allow mild hypercapnia (target $PaCO_2$ 45–50 mmHg, up to 60 mmHg if tolerated)
- ✓ Monitor pH, maintain >7.20
- ✓ Review contraindications: severe pulmonary hypertension, increased intracranial pressure, severe cardiac arrhythmias, pre-existing severe acidosis

**Perioperative Therapy Bundle**

- ✓ Early mobilization and pain control
- ✓ Respiratory physiotherapy postoperatively
- ✓ Consider high-flow oxygen therapy post-extubation in high-risk patients with hypoxemia
- ✓ Continue close monitoring of oxygenation, ventilation, and hemodynamics

**Limits of Physiological Gain**

- ✓ Before applying any physiological manipulation (increasing PEEP, alveolar recruitment maneuvers, high $F_IO_2$), consider the uncertainty of benefit for patient-centered outcomes
- ✓ Avoid excessive interventions that may not improve, or may even worsen patient outcomes
- ✓ Regularly reassess the risk-benefit ratio of each intervention

**FIG. 1** Checklist for protective one-lung ventilation. $PaCO_2$, arterial pressure of carbon dioxide; $SpO_2$, peripheral capillary oxygen saturation.

group (5.6 $cmH_2O$), yet this did not translate into improved clinical outcomes. The results are in line with an RCT with more than 1100 patients with thoracic surgery undergoing OLV showing that a PEEP titrated according to the lowest driving pressure did not reduce PPCs [6■]. This is in line with the results of a meta-analysis highlighting that while individualized PEEP titration reduced PPCs in certain subgroups – particularly when dynamic compliance or stepwise decremental strategies were used – the generalizability of these results is limited [7]. The benefit was less clear in patients with short surgery durations, mild preoperative lung injury, and when individualized PEEP was titrated by methods other than dynamic compliance. Furthermore, the same meta-analysis acknowledged that the inclusion of studies combining intraoperative and postoperative interventions, such as early mobilization and incentive spirometry, might have overestimated the benefits attributable solely to intraoperative individualized PEEP [7]. In this context, it is noteworthy that a randomized clinical trial of 880 patients undergoing major lung resection revealed no significant differences in terms of PPCs between patients ventilated with a tidal volume of 6 ml/kg of predicted body weight without PEEP and patients ventilated with a bundle based on a tidal volume of 4 ml/kg of predicted body weight, alveolar recruitment maneuvers, and fixed PEEP of 5 $cmH_2O$. This underscores the multifactorial etiology of complications and the complexity of isolating the most significant causal factors [12].

The iPROVE-OLV trial further emphasized the importance of the postoperative component in reducing PPCs, suggesting that the observed benefits may be because of the perioperative bundle of a lung expansion strategy rather than individualized PEEP alone [10■]. The limitations of recent meta-analyses are important to consider though. Heterogeneity in outcome definitions, PEEP titration methods, timing of measurements, and patient populations complicates the interpretation and generalizability of results. Many studies included in meta-analyses combined intraoperative and postoperative interventions, making it difficult to isolate the effect of individualized PEEP. In addition, the majority of trials excluded high-risk populations such as patients with severe obesity, significant lung or heart disease, or those undergoing emergency surgery, further limiting the applicability of findings to broader clinical practice.

The available evidence underscores that the debate on individualized PEEP remains unresolved. While individualized PEEP shows promise in select cases – particularly when paired with compliance-based titration and as part of a comprehensive perioperative bundle – the lack of consistent evidence for reduced PPCs across all patient populations necessitates further high-quality research. The potential benefits of individualized PEEP are context-dependent and may not apply broadly to all patients undergoing thoracic surgery. Furthermore, the studies on individualized PEEP have used fixed low-to-moderate levels of PEEP. Therefore, the question of whether individualized PEEP outperforms fixed higher PEEP levels remains unanswered.

Yet, the use of higher intraoperative PEEP levels to reduce PPCs in thoracic surgery remains a subject of considerable debate, with recent high-quality evidence challenging the routine use of such strategies. Large, contemporary RCTs and meta-analyses conducted have failed to demonstrate a consistent benefit of higher fixed PEEP levels – typically defined as 10 $cmH_2O$ or greater – over lower PEEP (5 $cmH_2O$) in reducing the incidence of PPCs during thoracic procedures requiring OLV [5,13]. The PROTHOR trial, the largest and most rigorously designed study to date on this question, directly compared high PEEP (10 $cmH_2O$ with recruitment maneuvers) to low PEEP (5 $cmH_2O$ without recruitment maneuvers) in over 2100 patients undergoing thoracic surgery with OLV [13]. The results showed no significant reduction in the composite incidence of PPCs in the high PEEP group, despite improved intraoperative oxygenation and compliance, and a higher rate of intraoperative hypotension was observed with the higher PEEP strategy. The application of higher PEEP is not without risk. Elevated PEEP levels can increase intrathoracic pressure, reduce venous return, and precipitate hemodynamic instability, particularly in patients with limited cardiovascular reserve or those undergoing lengthy or complex procedures. The evidence suggests that the potential harms of indiscriminately applying high PEEP may outweigh the modest and inconsistent benefits [14]. In summary, while higher intraoperative PEEP can improve oxygenation and reduce atelectasis in the short term, current high-quality evidence does not support its routine use as a necessary strategy to reduce PPCs in all patients. Instead, a moderate or individualized PEEP approach – tailored to patient physiology and intraoperative conditions – appears sufficient for most, with higher PEEP reserved for specific scenarios such as high risk of atelectasis, or demonstrated recruitability, always balanced against the risk of hemodynamic compromise.

## Driving pressure

Driving pressure, defined as plateau pressure minus PEEP, reflects the distending pressure applied to the lung and has been used as a surrogate for lung strain. Studies and meta-analyses have established driving pressure as a key modifiable risk factor for PPCs in thoracic surgery [7,10▪]. A recent meta-analysis found that individualized PEEP titration by compliance of the respiratory system reduced driving pressure by a mean of 2.13 $cmH_2O$ compared with fixed PEEP, correlating with reduced PPCs [7]. In a recent retrospective observational study of more than 3000 patients undergoing lung resection surgery, multivariable analysis revealed that each additional centimeter of water of intraoperative median driving pressure was associated with a 7% higher risk of impaired oxygenation [odds ratio (OR) 1.07, 95% CI: 1.04–1.10] [15]. Accordingly, a monocentric RCT demonstrated that driving pressure-guided ventilation during OLV reduced the incidence of PPCs compared with conventional protective ventilation [16]; however, not all studies have demonstrated a reduction in PPCs with driving pressure-guided ventilation, particularly when baseline driving pressures are already low [6▪]. Importantly, the optimal threshold for driving pressure remains debatable, but values below 13–15 $cmH_2O$ are generally considered protective [17]. Possibly, when low tidal volumes are used and the absolute level of driving pressure is above that threshold, avoiding increases in driving pressure plays a more important role.

## Mechanical power and energy

Mechanical power is a composite measure representing the total energy delivered to the respiratory system per unit time during mechanical ventilation [18]. It incorporates tidal volume, respiratory rate, airway pressures, and, according to some equations, even PEEP, and is thought to reflect the risk of ventilator-induced lung injury (VILI) more comprehensively than any single ventilatory parameter. Mechanical energy, in contrast, represents the cumulative energy delivered over the duration of ventilation [19■]. A 2024 prospective observational study in low-risk surgical patients found that higher intraoperative mechanical power was the sole significant predictor of PPCs, outperforming traditional variables such as tidal volume, PEEP, plateau pressure, and driving pressure [20]. Every 1 J/min increase in mechanical power was associated with 10% higher odds of PPCs, with a threshold of 12 J/min identified as likely hazardous. These findings were echoed in a 10-year retrospective cohort study of over 33 000 noncardiac, nonthoracic elective surgery patients, where increased intraoperative mechanical power was independently associated with higher odds of PPCs, ICU admission, longer hospital stay, and increased mortality [21]. In thoracic surgery, the picture, however, is more nuanced. During OLV, the ventilated lung is exposed to higher stress and strain due to reduced lung volume and altered mechanics, increasing the risk of VILI and PPCs; however, recent studies have challenged the predictive value of mechanical power for PPCs in the context of modern lung-protective ventilation strategies [22,23]. Recent studies demonstrate that mechanical power during OLV is not consistently associated with PPCs [22,23]. A prospective trial of 622 patients found no significant association between mechanical power and pulmonary complications, likely because of the implementation of standardized lung-protective ventilation strategies [23]. The median mechanical power in patients who developed PPCs was 6.76 J/min, notably lower than the threshold of 12 J/min identified in animal models for VILI [24]. In fact, a small subset of patients exceeded this threshold (>12 J/min), and among them, the incidence of PPCs was higher, suggesting that mechanical power may be more relevant in high-risk or prolonged cases. A post hoc analysis of a large RCT showed that mechanical energy was a stronger predictor of PPCs than mechanical power, especially in cases with prolonged duration of ventilation [6■,22]. This finding suggests that the cumulated mechanical energy may be more relevant to lung injury risk than the rate of energy delivery, particularly in longer or more complex surgeries. Similarly, a pediatric study in children less than or equal to 2 years undergoing video-assisted thoracic surgery found no significant association between mechanical power and PPCs, with other factors such as PEEP, duration of OLV, and surgeon experience being more predictive of adverse pulmonary events [25]. Finally, a registry-based cohort study involving 878 patients demonstrated that the adjusted odds of postoperative respiratory failure increased with each additional 1 J/min of mechanical power delivered during OLV. Analysis of the data revealed a U-shaped relationship between mechanical power and the risk of respiratory failure, identifying approximately 6.4 J/min as the optimal threshold associated with the lowest risk [26]. The variability in findings underscores the need for targeted application of mechanical power thresholds based on individual patient factors, surgical duration, and intraoperative events. The current evidence suggests that while mechanical power and energy are valuable concepts for understanding VILI, their utility as universal predictors of PPCs in thoracic surgery is limited by the complexity of patient and procedural factors. The adoption of lung-protective ventilation strategies – including low tidal volumes and careful monitoring of driving pressure – may attenuate the impact of mechanical power on clinical outcomes. Future research might aim to define patient-specific thresholds for mechanical power and energy, and to clarify their role in guiding intraoperative ventilation management.

## Chemical power

Chemical power in mechanical ventilation refers to the metabolic and biochemical effects of different ventilation strategies, particularly how oxygen and carbon dioxide management influence tissue oxygenation, acid-base balance, and cellular function [27]. This emerging concept is especially relevant during OLV, where maintaining optimal gas exchange is challenging.

Although the term chemical power is less frequently used in the literature compared with mechanical power, it captures the downstream physiological consequences of ventilatory decisions – such as permissive hypercapnia or hyperoxia – and their roles in inflammation, oxidative stress, and organ function [28]. In other words, the objective of chemical power is to address the biochemical and inflammatory components of lung damage. The elevated chemical power is manifested through several physiological responses, including an augmented tissue oxygen consumption, an overproduction of reactive oxygen species, a redox imbalance, and cytokine activation. In theory, excessive oxidative stress from reperfusion at the end of OLV could also increase chemical power and lung damage. Unlike mechanical power, for which quantifiable frameworks exist, chemical power remains a largely theoretical concept without robust, standardized clinical metrics. A recently proposed equation estimates the chemical power generated by superoxide production using lung oxygen consumption as a function of inspired oxygen fraction and the first electron affinity of oxygen [29]. This model suggests that $FiO_2$ levels greater than or equal to 90% which are often used during OLV, reach a threshold relevant to experimental hyperoxic lung injury, aligning with observed clinical risk [29]. In an individual patient data analysis of three randomized clinical trials of intraoperative two-lung ventilation including more than 2400 patients, an increase of 1 J/min in chemical power was associated with 8% higher odds of PPC (odds ratio: 1.08, 95% CI: 1.05–1.10) [30▪]. In a retrospective cohort study of 2936 lung resection patients with generally adequate intraoperative oxygenation (median $SpO_2 \geq 95\%$), a higher fraction of inspired oxygen ($FiO_2 \geq 0.8$) was associated with an increased risk of impaired postoperative oxygenation [31]; however, strong evidence linking these practices to PPCs in thoracic surgery is still lacking. In addition, concerns persist regarding the effects of these strategies on pulmonary vascular resistance and right heart function, particularly during OLV.

## Therapy bundles and postoperative care

Recent studies have reinforced the value of enhanced recovery after surgery (ERAS) protocols in thoracic surgery [5], demonstrating reductions in hospital length of stay, opioid consumption, and postoperative morbidity. A recent multicenter prospective cohort study found that higher compliance with ERAS elements was strongly associated with improved outcomes after lung resection, including lower 30-day morbidity and shorter hospital stays [32]. Key elements with the greatest impact included early mobilization, optimal pain control, and standardized chest tube management.

Furthermore, the postoperative management of thoracic surgical patients is increasingly incorporating noninvasive respiratory support strategies. Recent studies have indicated a significant increase in the utilization of high-flow nasal oxygen (HFNO) over the preceding decade [10▪,33]. This growth can be attributed to the physiological benefits associated with its use and its potential role in the mitigation of respiratory complications. HFNO therapy delivers heated, humidified oxygen at flow rates up to 60 L/min, providing several physiological benefits: improved oxygenation, low-level PEEP, dead space washout, reduced work of breathing, and enhanced mucociliary clearance. These effects are particularly relevant in the immediate postoperative period after thoracic surgery, when patients are at high risk for hypoxemia, atelectasis, and respiratory failure because of pain, reduced lung volumes, and impaired cough [33]. A recent systematic review and meta-analysis of five RCTs (564 patients) found that prophylactic HFNO after lung resection did not significantly reduce the incidence of postoperative hypoxemia compared with conventional oxygen therapy, but did improve the oxygenation index ($PaO_2/FiO_2$) within the first 12 h postextubation [34]. No significant differences were observed in reintubation rates, escalation of respiratory support, or hospital length of stay. In a recent trial involving 116 patients, preventive use of HFNO within 24 h following thoracic surgery did not reduce the incidence of postoperative respiratory complications compared with standard oxygen therapy [35]; however, in patients who developed postoperative diaphragmatic dysfunction, HFNO reduced PPCs. A retrospective study in 72 esophagectomy patients found that HFNO improved the respiratory oxygenation index and reduced respiratory rate 24 h after extubation compared with conventional oxygen therapy, with a trend toward lower rates of acute respiratory failure (0 versus 13%; $P = 0.06$) [36]. HFNO benefits may be most pronounced in high-risk patient subgroups, such as those with preexisting lung disease, obesity, or prolonged surgery, but further research is needed to identify these populations and establish standardized protocols for HFNO use [18]. Future trials should focus on identifying patient subgroups most likely to benefit from HFNO and on establishing standardized protocols for its use, including optimal flow rates, duration, and criteria for escalation or de-escalation of therapy.

## Open questions

Despite significant advances in our understanding of intraoperative ventilation and postoperative respiratory support, several key questions remain unresolved:

- PEEP titration methods: There is no consensus on the best method for individualized PEEP titration. Approaches based on compliance, driving pressure, distribution of ventilation or respiratory mechanics, and lung ultrasound variables have advantages and limitations each, and their comparative effectiveness in different patient populations and surgical contexts remains to be established.
- Fixed high PEEP: It is not known whether individually titrated PEEP outperforms fixed higher PEEP based on the population's characteristics and risks. Because no study conducted so far has performed such a comparison, the value of individualized PEEP remains unproved.
- Thresholds for mechanical power and energy:

 

The identification of clinically relevant thresholds for mechanical power and cumulative mechanical energy that predict PPCs in thoracic surgery is an area of active investigation. The interplay between mechanical power, ventilation duration, and patient-specific risk factors warrants further study

- HFNO protocols and patient selection: The optimal use of HFNO in the postoperative period – including patient selection, timing, and integration with other respiratory support modalities – requires further clarification through well-designed RCTs
- Integration of perioperative bundles: The relative contributions of intraoperative ventilation strategies, postoperative respiratory support, early mobilization, and other elements of perioperative care bundles to reducing PPCs still need to be determined. Future research should aim to isolate the effects of individual interventions and to identify synergistic combinations for high-risk patients.

## CONCLUSION

The management of intraoperative mechanical ventilation during thoracic surgery is characterized by ongoing controversy and evolving strategies. While individualized PEEP titration is physiologically reasonable and can improve intraoperative respiratory mechanics, recent high-quality evidence demonstrates that it does not universally reduce PPCs. The benefits of individualized PEEP appear to be context-dependent, with the greatest promise in select patient subgroups and when integrated into comprehensive perioperative care bundles, but have not been tested against fixed high levels of PEEP. Driving pressure is a simple and reliable proxy for lung strain, integrating tidal volume, and compliance into a single parameter. Furthermore, the role of mechanical power and energy as predictors of PPCs is nuanced, with cumulative energy delivery potentially more relevant than instantaneous power, particularly in prolonged or complex surgeries. HFNO offers advantages in postoperative oxygenation and patient comfort but has not consistently demonstrated a reduction in PPCs.

The current evidence underscores the importance of individualized, patient-centered approaches to intraoperative ventilation and postoperative care, tailored to the specific risk profile and physiological characteristics of each patient. Further high-quality research is needed to refine PEEP titration methods, define mechanical and chemical power thresholds, optimize HFNO protocols, and integrate these strategies into effective perioperative bundles. Until such evidence becomes available, clinicians should apply individualized strategies judiciously and in a context-specific manner. This should be guided by an ongoing assessment of patient needs and responses, as well as emerging best practices.

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